VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
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COMPUTER NETWORK REPORT

Lab 7

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1 Beacon Frames

1.1 Question 1

Question: What are the SSIDs of the two access points that are issuing most of the beacon frames in this trace?

Answer: The SSIDs of the two access points that are issuing most of the beacon frames:

- **Access Point 1:** SSID `"30 Munroe St"` (BSSID: `00:16:b6:f7:1d:51`) accounts for **718 beacon frames** (over 95% of all beacons).
- **Access Point 2:** SSID `"linksys12"` (BSSID: `00:06:25:67:22:94`) accounts for **25 beacon frames**.

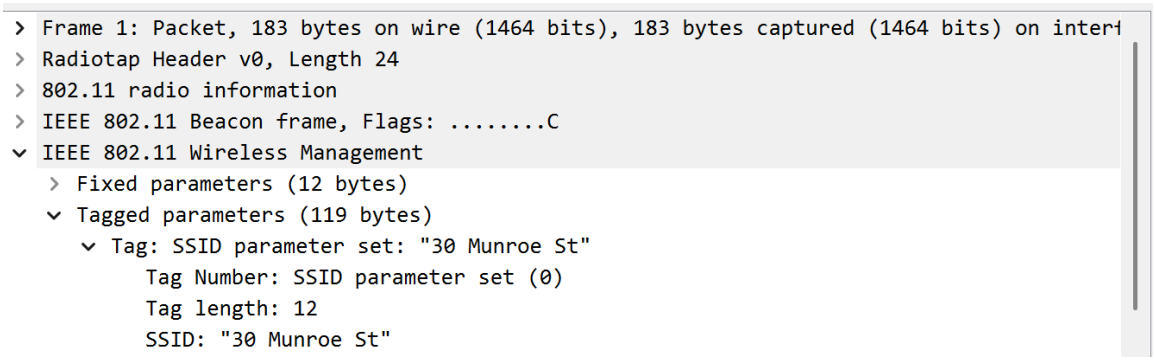


Figure 1.1

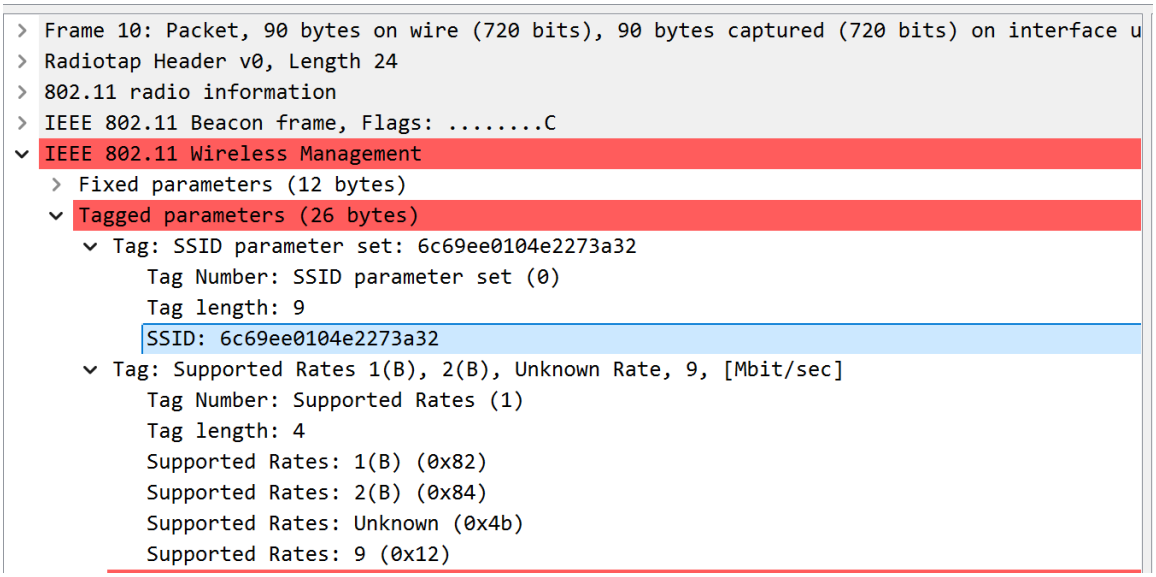


Figure 1.2

1.2 Question 2

Question: What are the intervals of time between the transmissions of the beacon frames from the *linksys_ses_24086* access point? From the *30 Munroe St.* access point?

Answer: The beacon interval is contained inside the fixed parameters field of the IEEE 802.11 management frame (`wlan.fixed.beacon_interval`). In the 802.11 standard, 1 Time Unit (TU) equals $1024\,\mu s = 0.001024\,s$:

- **30 Munroe St** (Frame 1): **0.102400 seconds** (100 Time Units).
- **linksys_ses_24086** (Frame 1774): **0.102400 seconds** (100 Time Units).

Both access points specify a target beacon transmission interval of **100 TU (102.4 ms)**, which is the standard default for 802.11 Wi-Fi networks.

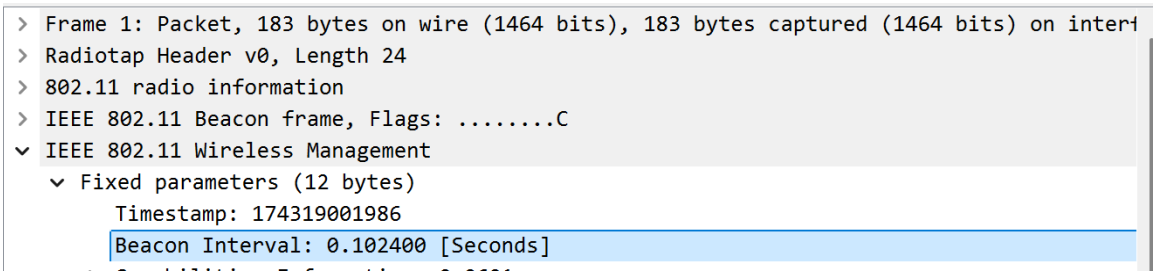


Figure 1.3

1.3 Question 3

Question: What (in hexadecimal notation) is the source MAC address on the beacon frame from 30 Munroe St?

Answer: The Transmitter/Source MAC address is specified in the **Address 2** field of the IEEE 802.11 MAC header:

- **Source MAC Address (Address 2 / SA):** `00:16:b6:f7:1d:51` (`CiscoLinksys_f7:1d:51`).

1.4 Question 4

Question: What (in hexadecimal notation) is the destination MAC address on the beacon frame from 30 Munroe St??

Answer: In Frame 1, the Destination/Receiver MAC address is located in the **Address 1** field of the IEEE 802.11 header:

- **Destination MAC Address (Address 1 / DA):** `ff:ff:ff:ff:ff:ff` (Broadcast).

Because beacon frames announce network presence to all nearby wireless stations in range, they are broadcasted to the universal broadcast address.

1.5 Question 5

Question: What (in hexadecimal notation) is the MAC BSS id on the beacon frame from 30 Munroe St?

Answer: In IEEE 802.11 infrastructure mode, the BSSID is the MAC address of the AP's wireless interface, contained in the **Address 3** field of the beacon frame:

- **MAC BSS ID (Address 3 / BSSID):** `00:16:b6:f7:1d:51` (`CiscoLinksys_f7:1d:51`).

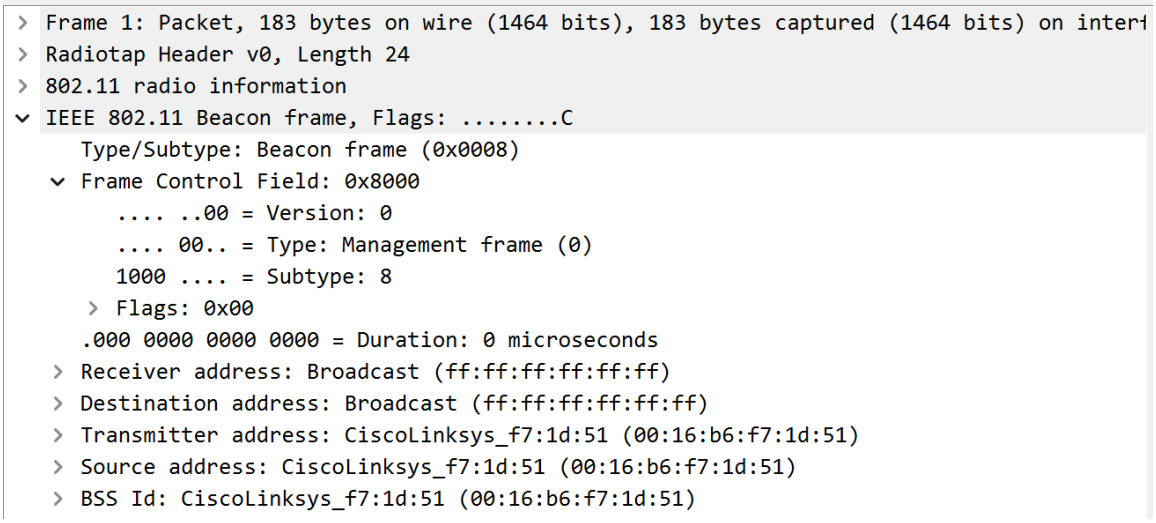


Figure 1.4

1.6 Question 6

Question: The beacon frames from the 30 Munroe St access point advertise that the access point can support four data rates and eight additional "extended supported rates." What are these rates?

Answer: Looking into the *Tagged parameters* section of Frame 1:

- **Supported Rates (Tag 1, 4 rates): 1.0, 2.0, 5.5, 11.0 Mbit/sec** (standard 802.11b DSSS basic rates).
- **Extended Supported Rates (Tag 50, 8 rates): 6.0, 9.0, 12.0, 18.0, 24.0, 36.0, 48.0, 54.0 Mbit/sec** (standard 802.11g OFDM rates).

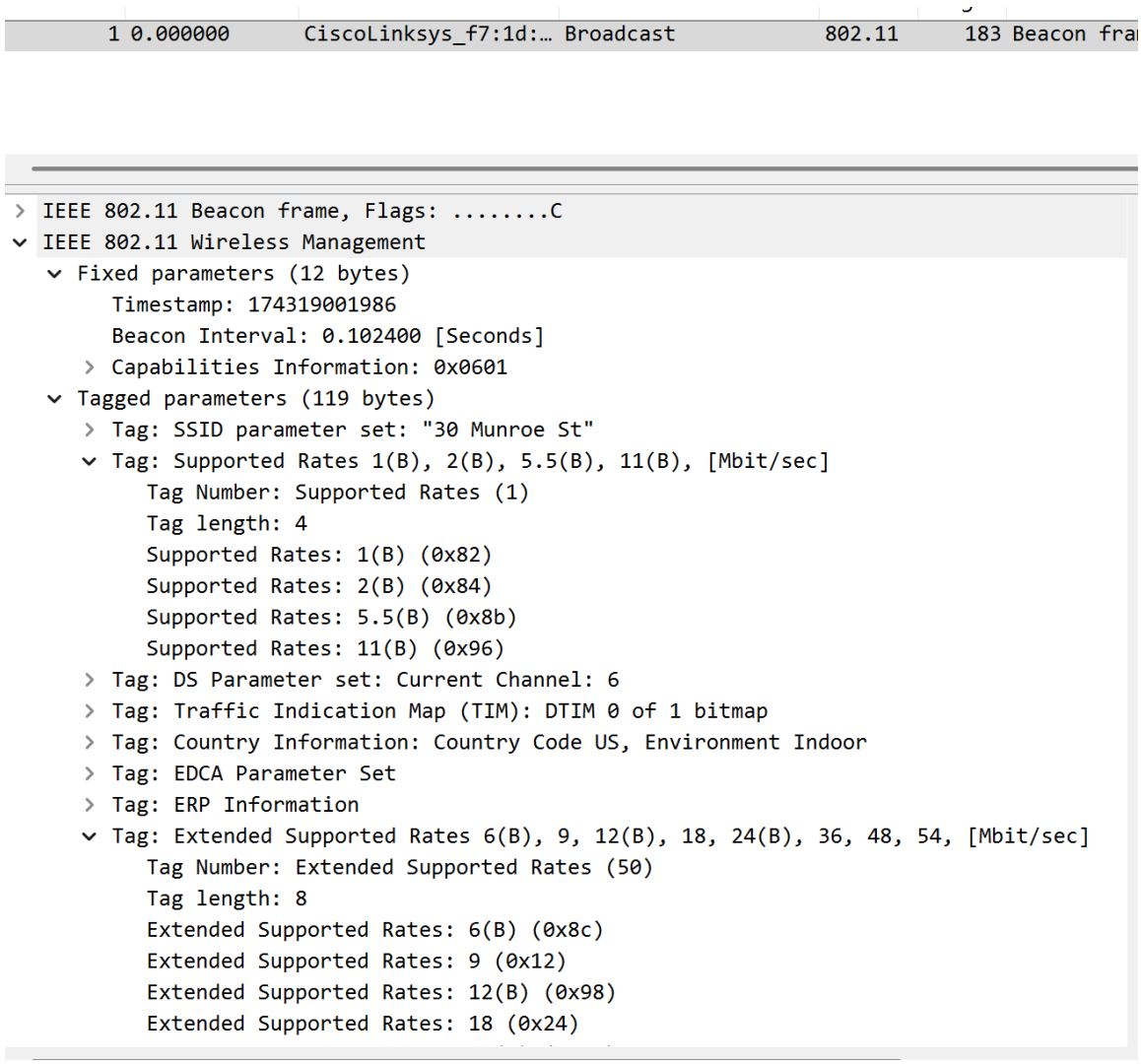

1 0.000000 CiscoLinksys_f7:1d:… Broadcast 802.11 183 Beacon fra
IEEE 802.11 Beacon frame, Flags: C
IEEE 802.11 Wireless Management
Fixed parameters (12 bytes)
Timestamp: 174319001986
Beacon Interval: 0.102400 [Seconds]
Capabilities Information: 0x0601
Tagged parameters (119 bytes)
Tag: SSID parameter set: "30 Munroe St"
Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
Tag Number: Supported Rates (1)
Tag length: 4
Supported Rates: 1(B) (0x82)
Supported Rates: 2(B) (0x84)
Supported Rates: 5.5(B) (0x8b)
Supported Rates: 11(B) (0x96)
Tag: DS Parameter set: Current Channel: 6
Tag: Traffic Indication Map (TIM): DTIM 0 of 1 bitmap
Tag: Country Information: Country Code US, Environment Indoor
Tag: EDCA Parameter Set
Tag: ERP Information
Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]
Tag Number: Extended Supported Rates (50)
Tag length: 8
Extended Supported Rates: 6(B) (0x8c)
Extended Supported Rates: 9 (0x12)
Extended Supported Rates: 12(B) (0x98)
Extended Supported Rates: 18 (0x24)

Figure 1.5

2 Data Transfer

2.1 Question 7

Question: Find the 802.11 frame containing the SYN TCP segment for this first TCP session (that downloads alice.txt). What are three MAC address fields in the 802.11 frame? Which MAC address in this frame corresponds to the wireless host (give the hexadecimal representation of the MAC address for the host)? To the access point? To the first-hop router? What is the IP address of the wireless host sending this TCP segment? What is the destination IP address? Does this destination IP address correspond to the host, access point, first-hop router, or some other network-attached device? Explain.

Answer: Looking at **Frame 474** ($t = 24.811093\,\text{s}$):

1. **Three MAC address fields in Frame 474:**
   - **Address 1 (Receiver Address - RA):** `00:16:b6:f7:1d:51` (`CiscoLinksys_f7:1d:51`) → Corresponds to the **Access Point (AP)**.
   - **Address 2 (Transmitter / Source Address - SA):** `00:13:02:d1:b6:4f` (`Intel_d1:b6:4f`) → Corresponds to the **Wireless Host**.
   - **Address 3 (Destination Address - DA):** `00:16:b6:f4:eb:a8` (`CiscoLinksys_f4:eb:a`) → Corresponds to the **First-hop router** (LAN/gateway interface).
2. **Wireless Host IP Address:** `192.168.1.109`.
3. **Destination IP Address:** `128.119.245.12` (`gaia.cs.umass.edu`).
4. **Explanation:** The destination IP address does **not** correspond to the host, the access point, or the first-hop router; it corresponds to an external remote web server hosting `alice.txt`. The wireless host addresses the link-layer frame to the AP (Address 1) to be forwarded to the first-hop router (Address 3), which decapsulates the link-layer header and routes the IP datagram across the wide-area Internet to the destination server.

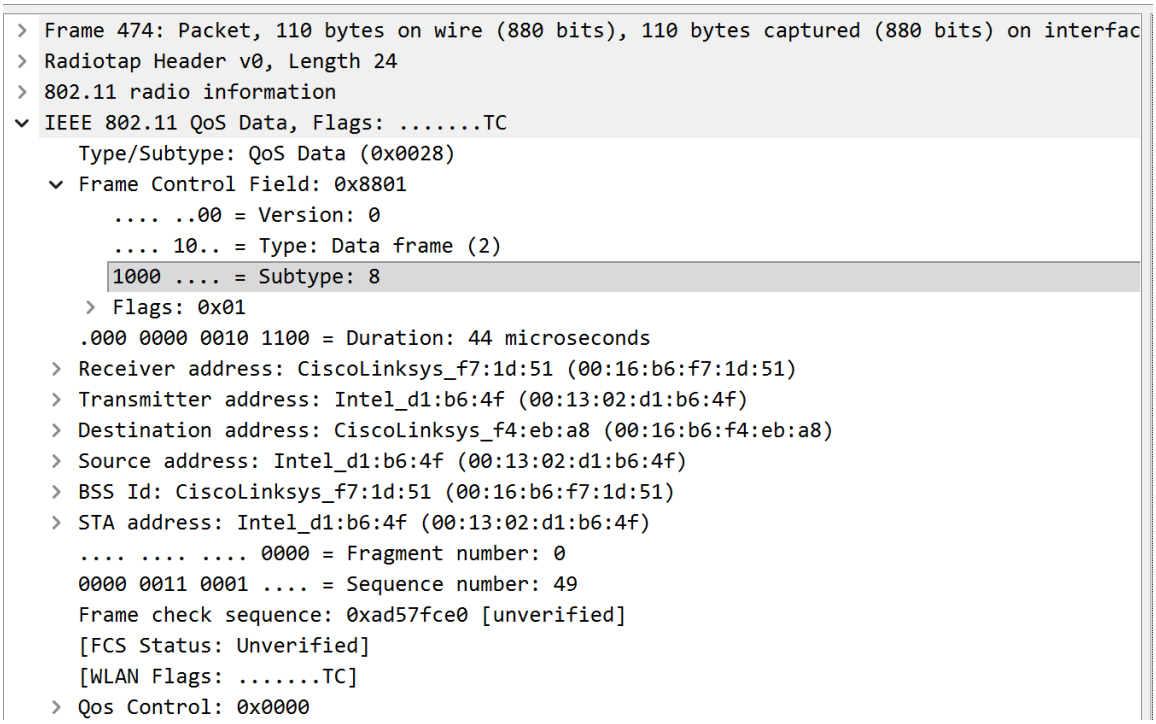

Frame 474: Packet, 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interfac
Radiotap Header v0, Length 24
802.11 radio information
IEEE 802.11 QoS Data, Flags: ....TC
Type/Subtype: QoS Data (0x0028)
Frame Control Field: 0x8801
.... ..00 = Version: 0
.... 10.. = Type: Data frame (2)
1000 .... = Subtype: 8
Flags: 0x01
.000 0000 0010 1100 = Duration: 44 microseconds
Receiver address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
Transmitter address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
Destination address: CiscoLinksys_f4:eb:a8 (00:16:b6:f4:eb:a8)
Source address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
STA address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
.... 0000 = Fragment number: 0
0000 0011 0001 .... = Sequence number: 49
Frame check sequence: 0xad57fce0 [unverified]
[FCS Status: Unverified]
[WLAN Flags: ....TC]
Qos Control: 0x0000

Figure 2.1

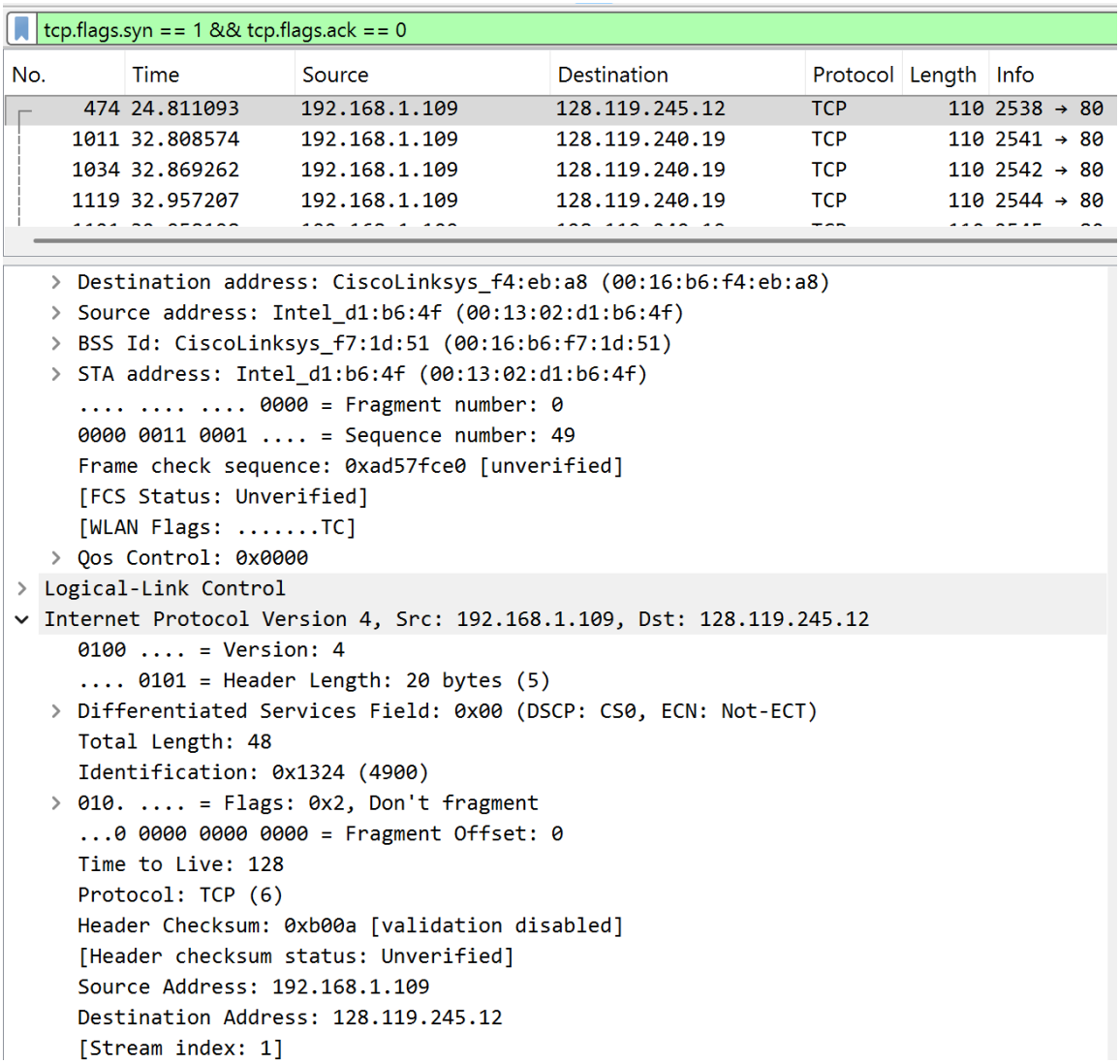


Figure 2.2

2.2 Question 8

Question: Find the 802.11 frame containing the SYNACK segment for this TCP session. What are three MAC address fields in the 802.11 frame? Which MAC address in this frame corresponds to the host? To the access point? To the first-hop router? Does the sender MAC address in the frame correspond to the IP address of the device that sent the TCP segment encapsulated within this datagram?

Answer: Looking at **Frame 476** ($t = 24.827751$ s):

1. **Three MAC address fields in Frame 476:**
   - **Address 1 (Receiver / Destination Address - RA/DA):** `91:2a:b0:49:b6:4f`

→ Corresponds to the **Wireless Host** (intended for host `00:13:02:d1:b6:4f`; captured with bit errors in the first 4 bytes over the air while retaining the lower bytes `b6:4f`).

- **Address 2 (Transmitter Address - TA):** `00:16:b6:f7:1d:51` (`CiscoLinksys_f7:1d:5`
  → Corresponds to the **Access Point (AP)**.
- **Address 3 (Source Address - SA):** `00:16:b6:f4:eb:a8` (`CiscoLinksys_f4:eb:a8`)
  → Corresponds to the **First-hop router**.

2. **Sender MAC address correspondence: No**, the sender MAC address (`00:16:b6:f4:eb:a8`) does **not** correspond to the IP address of the device that sent the TCP segment (`128.119.245.12`, the remote server). MAC addresses are local link-layer addresses valid only on the local network hop. The first-hop router received the IP datagram from the WAN and substituted its own MAC address as the source MAC before injecting it into the wireless BSS.

```
> Frame 476: Packet, 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interf
> Radiotap Header v0, Length 24
> 802.11 radio information
v IEEE 802.11 QoS Data, Flags: ..mP..F.C
    Type/Subtype: QoS Data (0x0028)
  v Frame Control Field: 0x8832
      .... ..00 = Version: 0
      .... 10.. = Type: Data frame (2)
      1000 .... = Subtype: 8
    > Flags: 0x32
    Duration/ID: 11560 (reserved)
  > Receiver address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
  > Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > Destination address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
  > Source address: CiscoLinksys_f4:eb:a8 (00:16:b6:f4:eb:a8)
  > BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > STA address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
```

Figure 2.3

3 Association and Disassociation

3.1 Question 9

Question: What two actions are taken (i.e., frames are sent) by the host in the trace just after t=49, to end the association with the 30 Munroe St AP that was initially in place when trace collection began? (Hint: one is an IP-layer action, and one is an 802.11-layer action). Looking at the 802.11 specification, is there another frame that you might have expected to see, but don't see here?

Answer: Right after $t = 49\,\text{s}$, the host performs two distinct actions to terminate its session:

- **IP-layer action (Frame 1733, $t = 49.583615\,\text{s}$):** The host transmits a **DHCP Release** message to release its leased IP address (`192.168.1.109`) back to the DHCP server/gateway (`192.168.1.1`).
- **802.11-layer action (Frame 1735, $t = 49.609617\,\text{s}$):** The host sends an **802.11 Deauthentication frame** (`wlan.fc.subtype == 12`) directly to the *30 Munroe St* AP (`00:16:b6:f7:1d:51`).

Missing frame from 802.11 specification: Under normal 802.11 procedures, a station ending an association is expected to send a **Disassociation frame** (`subtype 10`). However, in this trace, the station immediately sends a **Deauthentication frame** (`subtype 12`), which abruptly terminates both authentication and association in a single step without first disassociating.

No.	Time	Source	Destination	Protocol	Length	Info
1732	49.542481	CiscoLinksys_f7:1d:…	Broadcast	802.11	183	Beacon frame, SN=3588, FN=0, Flags=........C, BI=100, SSID="30 Munroe St"
1733	49.583615	192.168.1.109	192.168.1.1	DHCP	390	DHCP Release - Transaction ID 0xea5a526
1734	49.583771		Intel_d1:b6:4f	802.11	38	Acknowledgement, Flags=........C
1735	49.609617	Intel_d1:b6:4f	CiscoLinksys_f7:1d:…	802.11	54	Deauthentication, SN=1605, FN=0, Flags=........C

Figure 3.1

3.2 Question 10

Question: Examine the trace file and look for AUTHENTICATION frames sent from the host to an AP and vice versa. How many AUTHENTICATION messages are sent from the wireless host to the linksys_ses_24086 AP (which has a MAC address of Cisco_Li_f5:ba:bb) starting at around t=49?

Answer: The host transmits exactly **6 AUTHENTICATION messages** to the *linksys_ses_24086* AP:

- Frame 1740 ($t = 49.638857\,\text{s}$)
- Frame 1741 ($t = 49.639700\,\text{s}$)
- Frame 1742 ($t = 49.640702\,\text{s}$)
- Frame 1744 ($t = 49.642315\,\text{s}$)
- Frame 1746 ($t = 49.645319\,\text{s}$)
- Frame 1749 ($t = 49.649705\,\text{s}$)

The host sends an initial authentication request followed by 5 rapid retransmissions due to the lack of an authentication reply from the target AP.

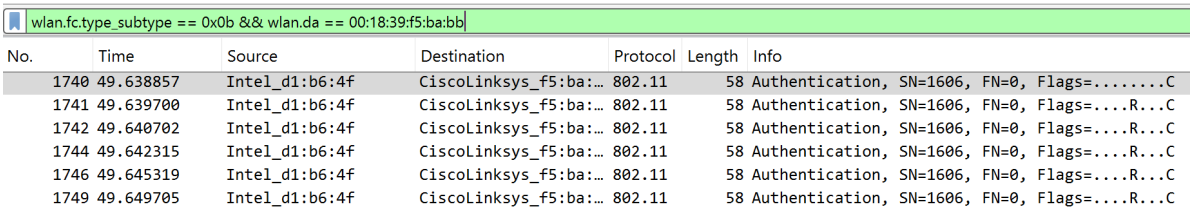


Figure 3.2

3.3 Question 11

Question: Does the host want the authentication to require a key or be open?
Answer: Looking at the Fixed Parameters field in Frame 1740:

- **Authentication Algorithm: Open System (0)**.

The host requests **Open System authentication**, meaning it does not require a shared key at the 802.11 link-layer authentication phase.

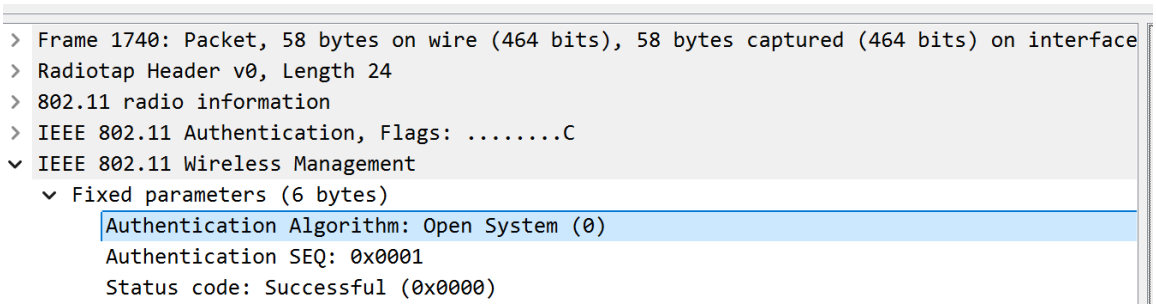


Figure 3.3

3.4 Question 12

Question: Do you see a reply AUTHENTICATION from the linksys_ses_24086 AP in the trace?

Answer: No. While the AP sends physical layer 802.11 ACK control frames, it never returns an 802.11 Management Authentication reply frame (0 reply frames present in the trace).

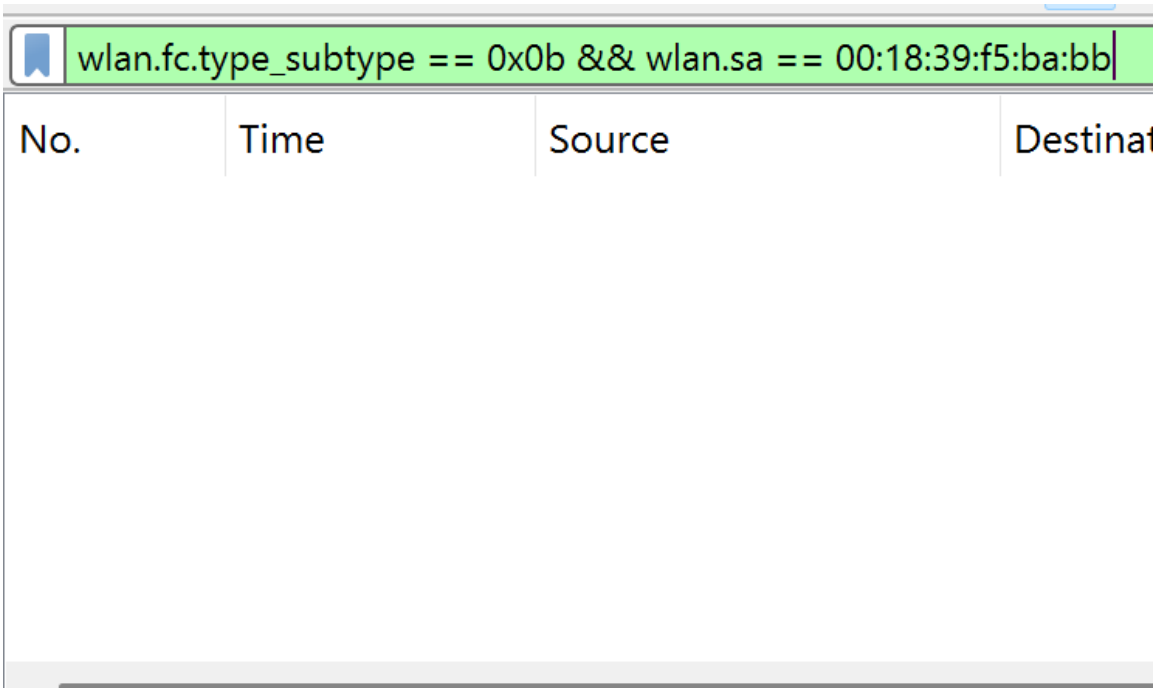


Figure 3.4

3.5 Question 13

Question: Now let's consider what happens as the host gives up trying to associate with the linksys_ses_24086 AP and now tries to associate with the 30 Munroe St AP. Look for AUTHENTICATION frames sent from the host to an AP and vice versa. At what times are there an AUTHENTICATION frame from the host to the 30 Munroe St. AP, and when is there a reply AUTHENTICATION sent from that AP to the host in reply?

Answer: Around $t \approx 63\,\text{s}$, the authentication frames exchanged between the wireless host and the *30 Munroe St* AP are:

- **Host → 30 Munroe St AP (Authentication Request):**
  - **Frame 2156** at **t = 63.168087 seconds** (Sequence number 1, Open System).
  - **Frame 2160** at **t = 63.169707 seconds** (Retransmission).
- **30 Munroe St AP → Host (Authentication Reply):**

- **Frame 2158** at **t = 63.169071 seconds** (Status: Successful).
- **Frame 2164** at **t = 63.170692 seconds** (Status: Successful).

```
2156 63.168087    Intel_d1:b6:4f       CiscoLinksys_f7:1d:… 802.11      58 Authentication, SN=1647, FN=0, Flags=........C
2158 63.169071    CiscoLinksys_f7:1d:… Intel_d1:b6:4f       802.11      58 Authentication, SN=3726, FN=0, Flags=........C
2160 63.169707    Intel_d1:b6:4f       CiscoLinksys_f7:1d:… 802.11      58 Authentication, SN=1647, FN=0, Flags=....R...C
2164 63.170692    CiscoLinksys_f7:1d:… Intel_d1:b6:4f       802.11      58 Authentication, SN=3727, FN=0, Flags=........C
```

Figure 3.5

3.6 Question 14

Question: An ASSOCIATE REQUEST from host to AP, and a corresponding ASSOCIATE RESPONSE frame from AP to host are used for the host to associate with an AP. At what time is there an ASSOCIATE REQUEST from host to the 30 Munroe St AP? When is the corresponding ASSOCIATE REPLY sent?

Answer: The times at which the association frames are exchanged:

- **ASSOCIATE REQUEST (Host → AP): Frame 2162** at **t = 63.169910 seconds**.
- **ASSOCIATE RESPONSE (AP → Host): Frame 2166** at **t = 63.192101 seconds** (Status: Successful).

```
2162 63.169910    Intel_d1:b6:4f       CiscoLinksys_f7:1d:… 802.11      89 Association Request, SN=1648, FN=0, Flags=........C, SSID="30 Munroe St"
2166 63.192101    CiscoLinksys_f7:1d:… Intel_d1:b6:4f       802.11      94 Association Response, SN=3728, FN=0, Flags=........C
```

Figure 3.6

3.7 Question 15

Question: What transmission rates is the host willing to use? The AP? To answer this question, you will need to look into the parameters fields of the 802.11 wireless LAN management frame.

Answer: Examining the *Tagged parameters* in the Association Request (**Frame 2162**) and Association Response (**Frame 2166**):

- **Host Transmission Rates (Frame 2162):**
  - *Supported Rates (Tag 1):* 1.0, 2.0, 5.5, 11.0, 6.0, 9.0, 12.0, 18.0 Mbit/sec.
  - *Extended Supported Rates (Tag 50):* 24.0, 36.0, 48.0, 54.0 Mbit/sec.
  - *Combined Rates:* **1.0, 2.0, 5.5, 6.0, 9.0, 11.0, 12.0, 18.0, 24.0, 36.0, 48.0, 54.0 Mbit/sec**.

- **AP Transmission Rates (Frame 2166):**
  - *Supported Rates (Tag 1):* 1.0, 2.0, 5.5, 11.0 Mbit/sec.
  - *Extended Supported Rates (Tag 50):* 6.0, 9.0, 12.0, 18.0, 24.0, 36.0, 48.0, 54.0 Mbit/sec.
  - *Combined Rates:* **1.0, 2.0, 5.5, 6.0, 9.0, 11.0, 12.0, 18.0, 24.0, 36.0, 48.0, 54.0 Mbit/sec**.

Both the host and the AP agree to support the exact same set of **12 data rates** (spanning 802.11b and 802.11g standards).

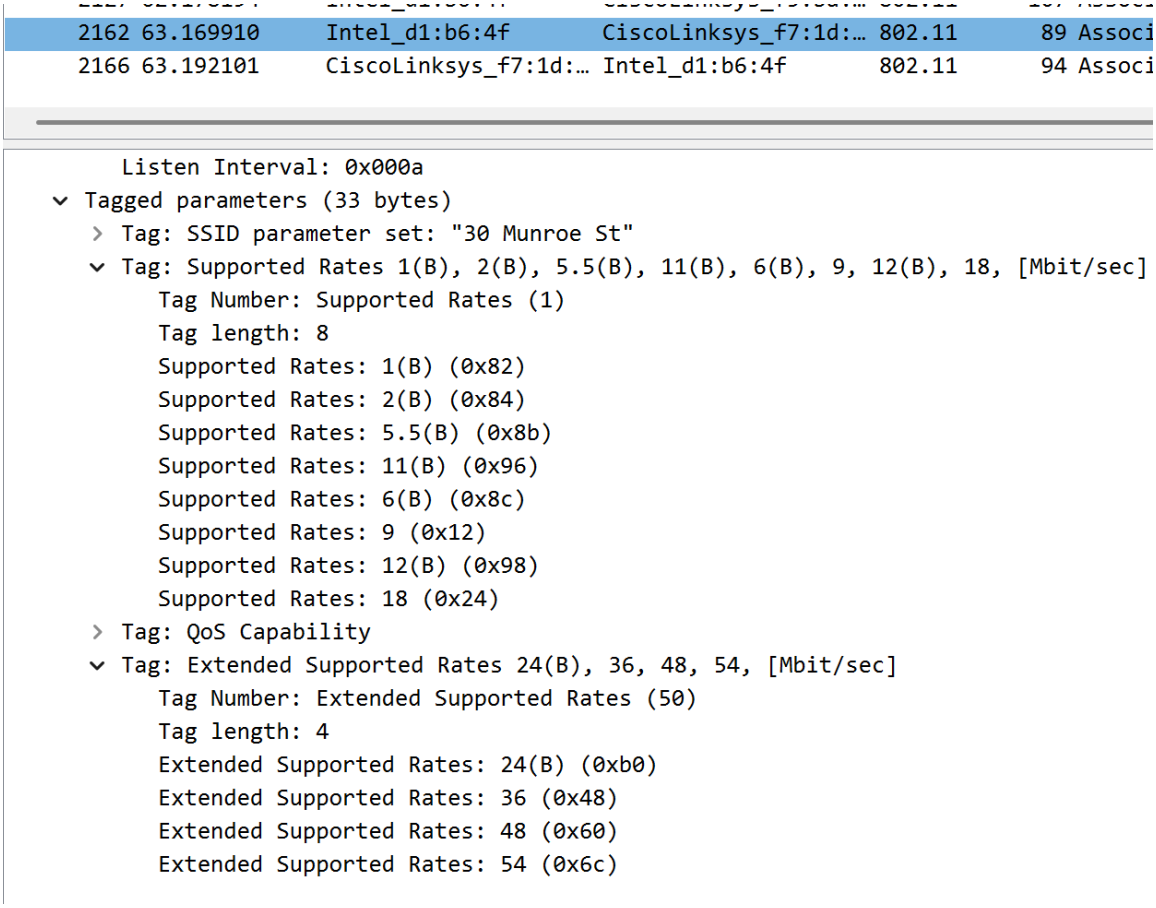


Figure 3.7

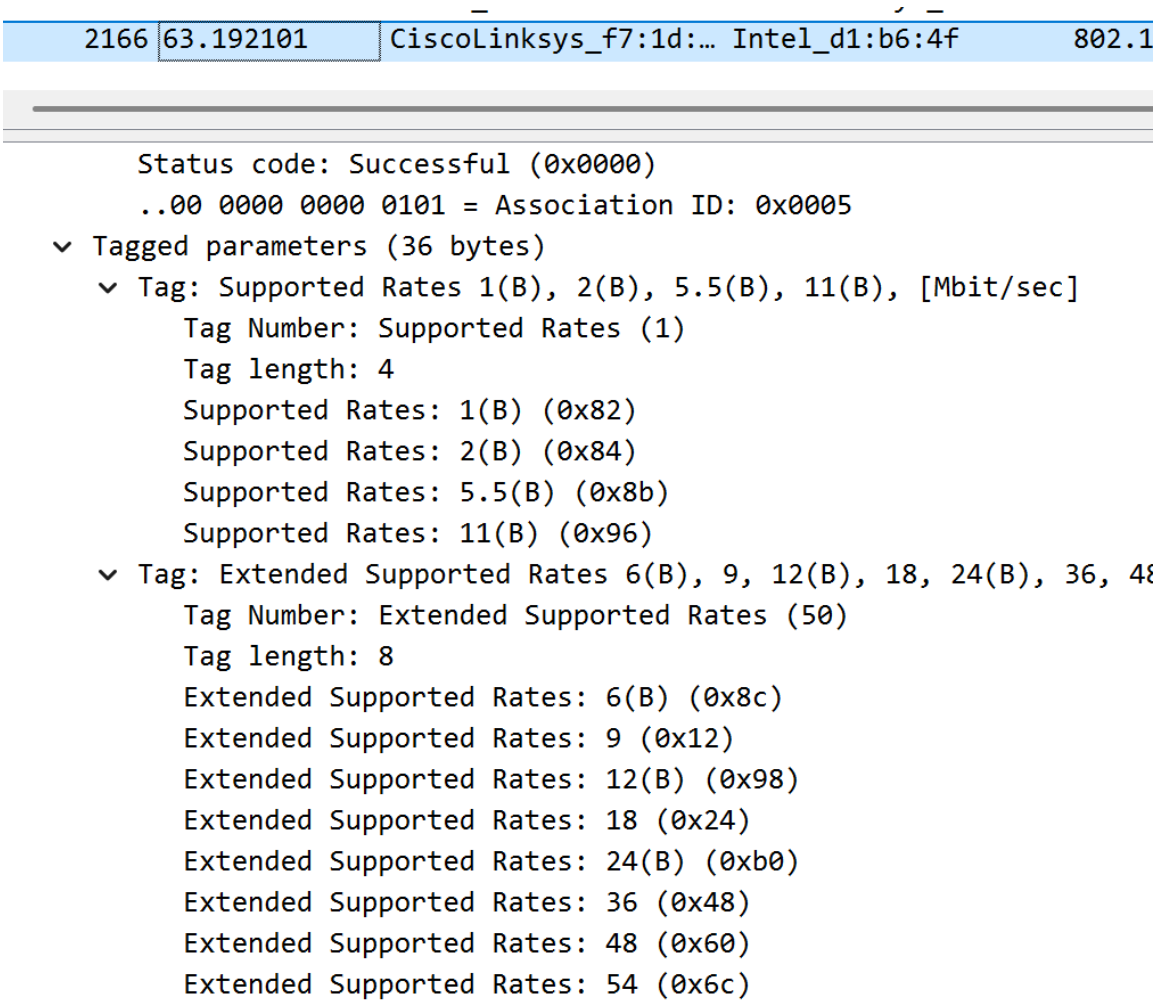

2166 63.192101 CiscoLinksys_f7:1d:… Intel_d1:b6:4f 802.1
Status code: Successful (0x0000)
..00 0000 0000 0101 = Association ID: 0x0005
Tagged parameters (36 bytes)
Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
Tag Number: Supported Rates (1)
Tag length: 4
Supported Rates: 1(B) (0x82)
Supported Rates: 2(B) (0x84)
Supported Rates: 5.5(B) (0x8b)
Supported Rates: 11(B) (0x96)
Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48
Tag Number: Extended Supported Rates (50)
Tag length: 8
Extended Supported Rates: 6(B) (0x8c)
Extended Supported Rates: 9 (0x12)
Extended Supported Rates: 12(B) (0x98)
Extended Supported Rates: 18 (0x24)
Extended Supported Rates: 24(B) (0xb0)
Extended Supported Rates: 36 (0x48)
Extended Supported Rates: 48 (0x60)
Extended Supported Rates: 54 (0x6c)

Figure 3.8

4 Other Frame Types

4.1 Question 16

Question: Our trace contains a number of PROBE REQUEST and PROBE RESPONSE frames. What are the sender, receiver and BSS ID MAC addresses in these frames? What is the purpose of these two types of frames?

Answer:

1. **Probe Request Frame (Frame 1737):**
   - **Receiver MAC (Address 1 / DA):** `ff:ff:ff:ff:ff:ff` (Broadcast).
   - **Transmitter / Sender MAC (Address 2 / SA):** `00:13:02:d1:b6:4f` (`Intel_d1:b6:4f`).
   - **BSS ID (Address 3):** `ff:ff:ff:ff:ff:ff` (Wildcard BSSID).
   - **Purpose:** Active network scanning. Wireless stations broadcast Probe Requests across radio channels to proactively discover available access points and query their parameters without passively waiting for beacon frames.
2. **Probe Response Frame (Frame 27):**
   - **Receiver MAC (Address 1 / DA):** `00:13:02:d1:b6:4f` (`Intel_d1:b6:4f`).
   - **Transmitter / Sender MAC (Address 2 / SA):** `00:16:b6:f7:1d:51` (`CiscoLinksys_f7:1d:51`).
   - **BSS ID (Address 3):** `00:16:b6:f7:1d:51` (`CiscoLinksys_f7:1d:51`).
   - **Purpose:** Sent by an AP as a unicast response to a Probe Request. It supplies the station with full network parameters (SSID, supported data rates, capability info) identical to a beacon, allowing the station to evaluate link quality and decide whether to associate.

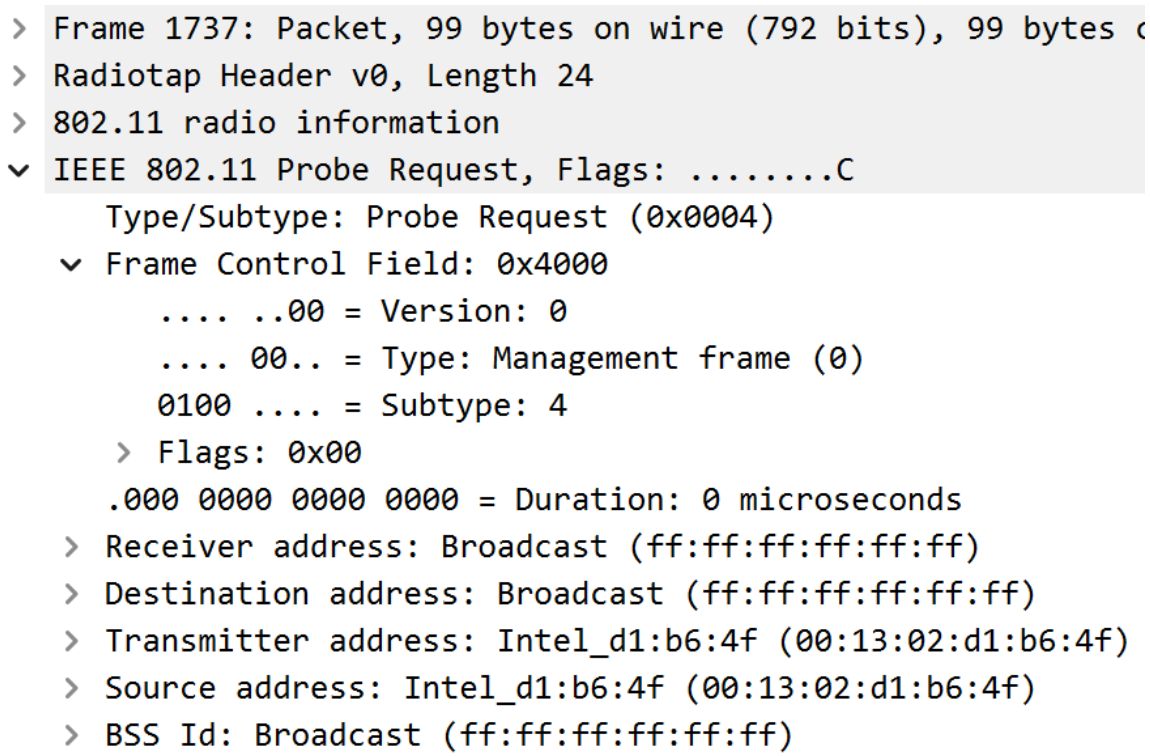


Figure 4.1

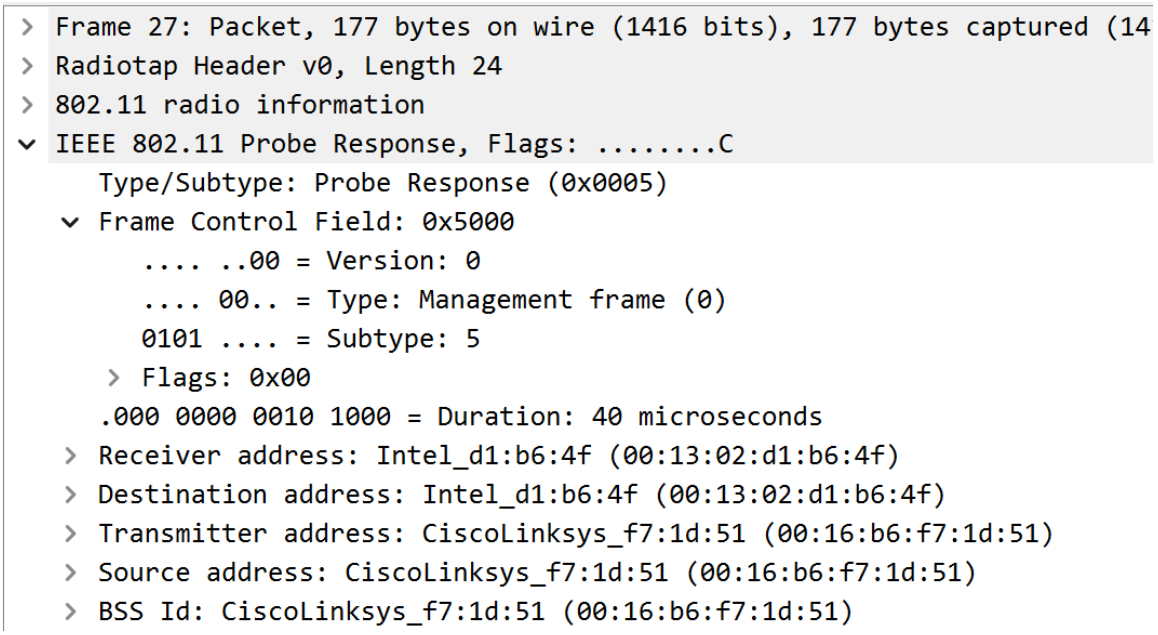


Figure 4.2